

Dhaka International University

New Building, Satarkul, Badda, Dhaka - 1212



Department Of CSE

ASSIGNMENT

Course Name : Digital Logic Design

Course Code : 0713-203

Assignment No : 01

Assignment On : Theory part

Submitted By:

Name : Md Murshid Mahmud

Roll No : 02

Reg. No : CS-E-118-24-127822

Batch : 118

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Submitted To:

Name: Md Maruf Ahmed

(Lecturer)

Dept. of CSE

Dhaka International University,
Dhaka.

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Assignment (Theory)

1. Mention different data error detection and correction techniques.

Ans: Error Detection Techniques:

- ① Parity check
- ② Checksum
- ③ Cyclic Redundancy check (CRC)
- ④ Lengthwise Redundancy check (LRC)

Error Correction Techniques:

- ① Hamming Code
- ② Forward Error Correction
- ③ Automatic Repeat request.

2. State and prove De Morgan's Theorems using truth tables.

Ans: There are two basic theorems of great importance in Boolean Algebra, which are De Morgan's First Law and De Morgan's Second Law. These are also called De Morgan's Theorems.

① De Morgan's First Law:

$$(P \cdot Q)' = (P)' + (Q)'$$

Truth Table:

P	Q	P'	Q'	(P · Q)'	(P)' + (Q)'
1	1	0	0	0	0
1	0	0	1	1	1
0	1	1	0	1	1
0	0	1	1	1	1

② De Morgan's Second Law:

$$(P + Q)' = (P)' \cdot (Q)'$$

Truth Table:

P	Q	P'	Q'	(P + Q)'	(P)' · (Q)'
1	1	0	0	0	0
1	0	0	1	0	0
0	1	1	0	0	0
0	0	1	1	1	1

3. Convert the binary number $(1101001)_2$ to

* Gray Code

* Excess-3 Code

* Binary to Gray code:

$$(1101001)_2 = (?)_{\text{Gray}}$$

$$\begin{array}{ccccccc} 1 & 1 & 0 & 1 & 0 & 0 & 1 \\ B_7 & B_6 & B_5 & B_4 & B_3 & B_2 & B_1 \end{array}$$

$$G_7 = B_7 \Rightarrow 1$$

$$G_6 = B_7 \oplus B_6 \Rightarrow 1 \oplus 1 = 0$$

$$G_5 = B_6 \oplus B_5 \Rightarrow 1 \oplus 0 = 1$$

$$G_4 = B_5 \oplus B_4 \Rightarrow 0 \oplus 1 = 1$$

$$G_3 = B_4 \oplus B_3 \Rightarrow 1 \oplus 0 = 1$$

$$G_2 = B_3 \oplus B_2 \Rightarrow 0 \oplus 0 = 0$$

$$G_1 = B_2 \oplus B_1 \Rightarrow 0 \oplus 1 = 1$$

$$(1101001)_2 = (10111101)_{\text{Gray}}$$

* Binary to Excess-3 Code:

$$(1101001)_2 = (?)_{\text{Excess-3}}$$

Step: 1/ $(1101001)_2 = (105)_{10}$

Step: 2/ $1+3=4$, $0+3=3$, $5+3=8$

Step: 3/ $(4)_{10} = (0100)_2$, $(3)_{10} = (0011)_2$, $(8)_{10} = (1000)_2$

$\therefore$ Excess-3 code is $\rightarrow (0100\ 0011\ 1000)_{\text{Excess-3}}$

4. Design SR Latches with operations.

Ans: SR Latch is a circuit with:

- ① 2 cross-coupled NAND gate
- ② 2 input S for SET and R for RESET
- ③ 2 output Q, Q'

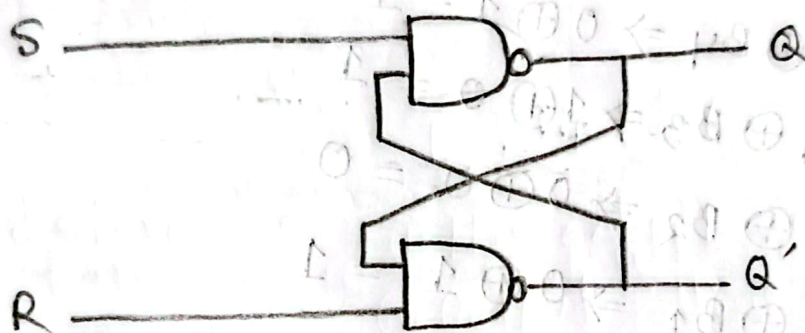


Figure: SR Latch

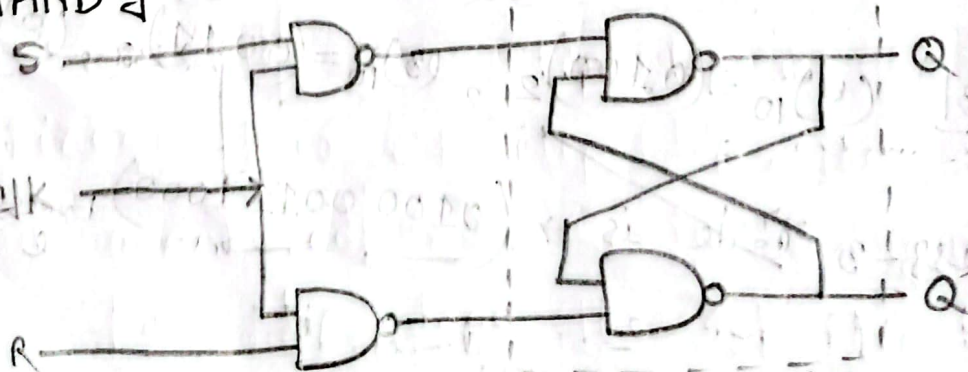
5. Design Flip Flops : SR, JK, Master slave, D and T

① SR FlipFlop:

Construction of SR Flip Flop by using NAND Latch

① NAND Latch

② Two NAND gates.



NAND Latch

ii) JK Flip Flop:

- ⊗ SR Flip Flop constructed from NAND latch.
- ⊗ Two other connections.

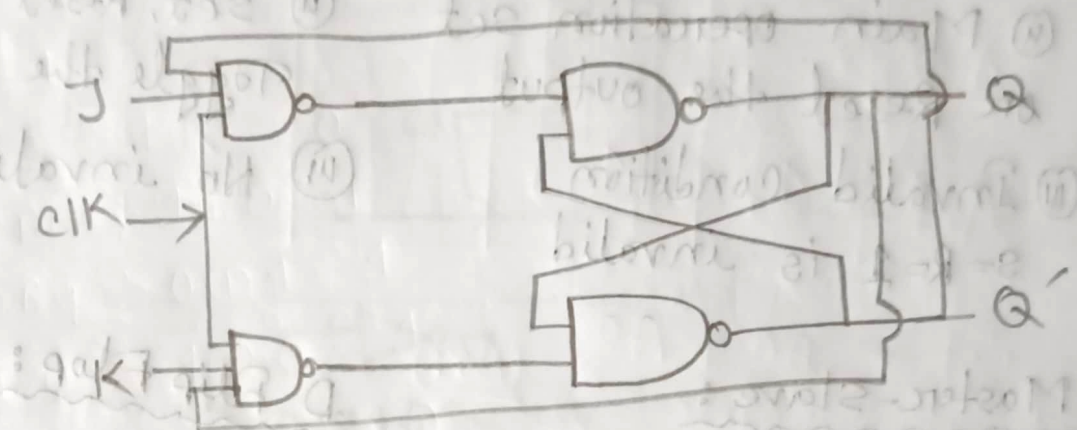
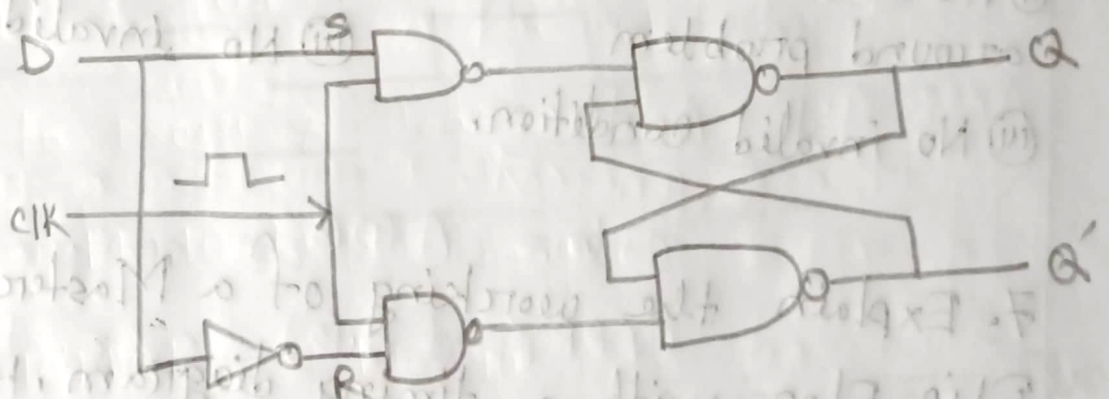
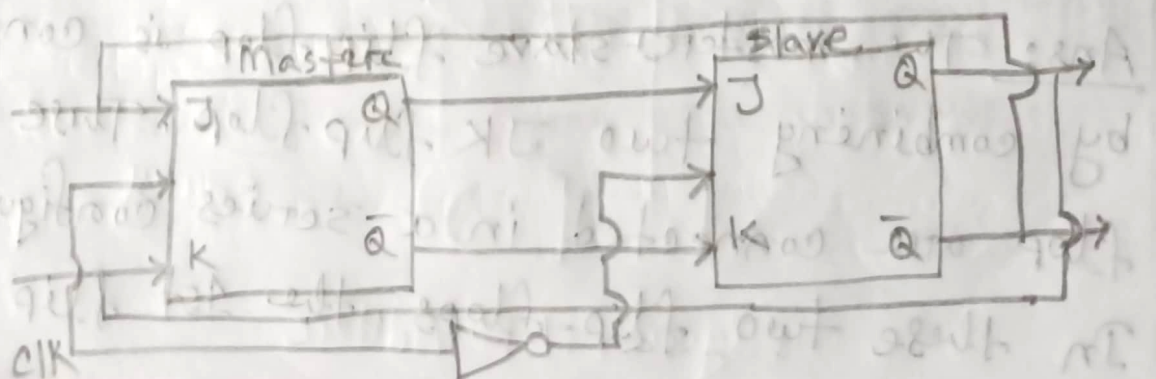


Figure: JK Flip Flop

iii) D Flip Flop:



iv) Master-Slave JK Flip Flop:



6. Compare SR, JK, Master Slave and D Flip Flop

Ans: SR:

- ① input has S and R
- ② Main operation set or Reset the output
- ③ Invalid condition $S=R=1$ is invalid

JK:

- ① J & K input
- ② Set, Reset and Toggle the output
- ③ No invalid condition

Master-Slave:

- ① Master and Slave Section
- ② Avoid the race around problem
- ③ No invalid condition.

D Flip Flop:

- ① Has only D input
- ② Output follow the input D
- ③ No invalid condition

7. Explain the working of a Master-Slave JK Flip-Flop with a timing diagram. How does it overcome race-around condition?

Ans: The master-slave flip flop is constructed by combining two JK flip flop. These flip flop are connected in a series configuration. In these two flip-flops, the 1st flip flop

work as "master", called the master flip flop and 2nd work as a "slave", called slave flip flop.

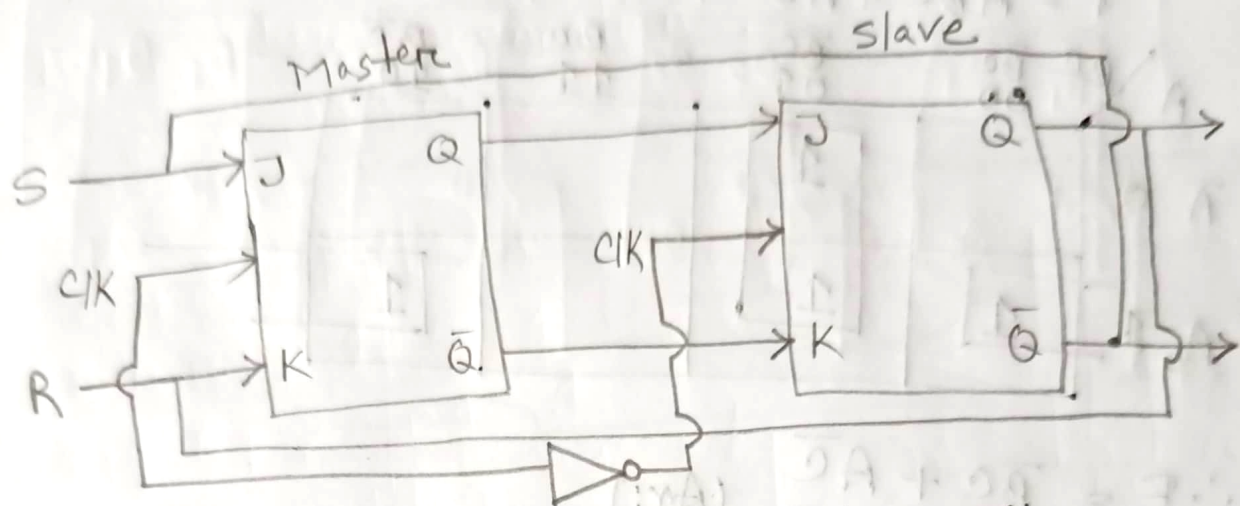


Figure: Master slave JK

Timing Diagram:



The race around condition occurs in a JK flip flop, when $J=K=1$ and the clock pulse remains active for a long time. The output may toggle repeatedly during the same clock pulse.

8. Solve using Kmap: $A'B'C + AB'C + AB'C' + ABC'$

Ans:

$$F = A'B'C + AB'C + AB'C' + ABC'$$

A \ BC	$\overline{B}\overline{C}$ 00	$\overline{B}C$ 01	$B\overline{C}$ 11	BC 10
$\overline{A}$ 0		1		
A 1	1	1		1

$$\therefore F = \overline{B}C + A\overline{C} \quad (\text{Ans:})$$

9. Design a Half Adder and Full Adder using two Half Adders.

Ans: Half Adder:

$$\text{Sum} = A \oplus B$$

$$\text{CARRY} = (A \cdot B)$$

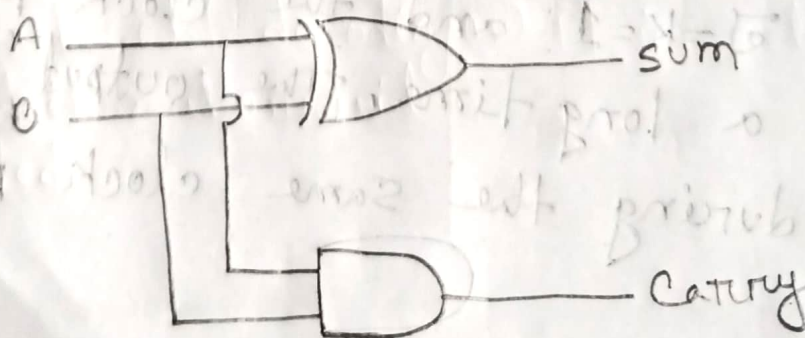


Figure: Half Adder

Full Adder:

$$\text{sum} = A \oplus B \oplus \text{Cin}$$

$$\text{Cout} = \text{Cin} (A \oplus B) + AB$$

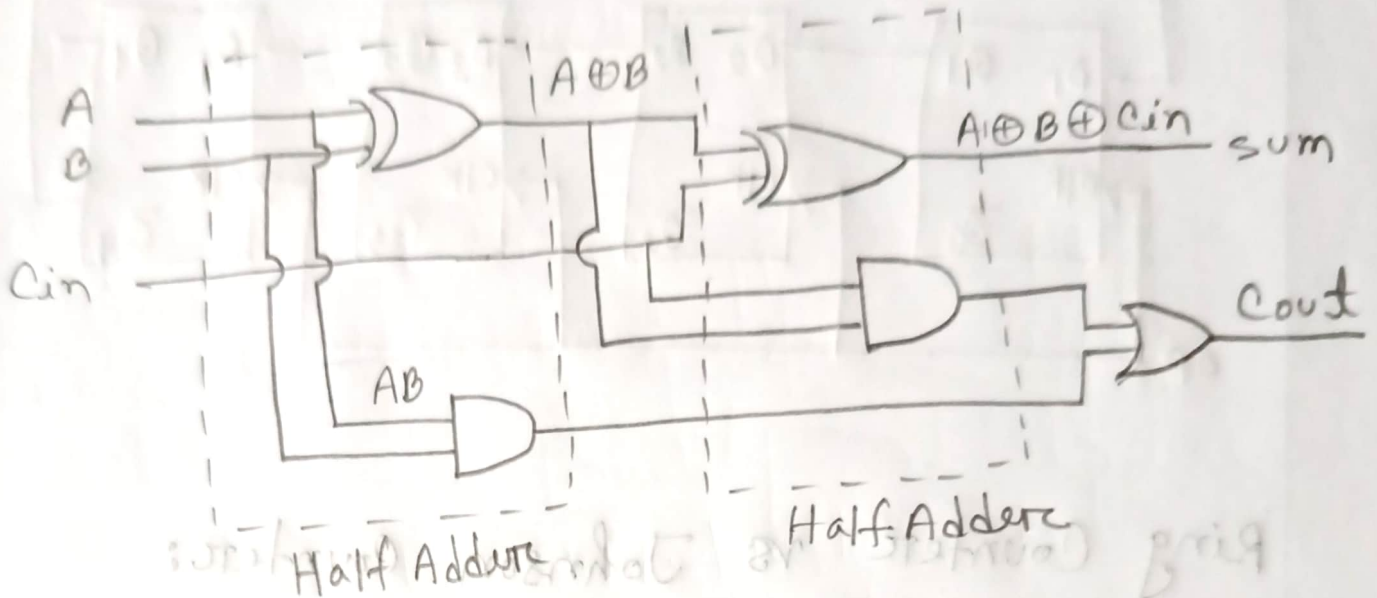
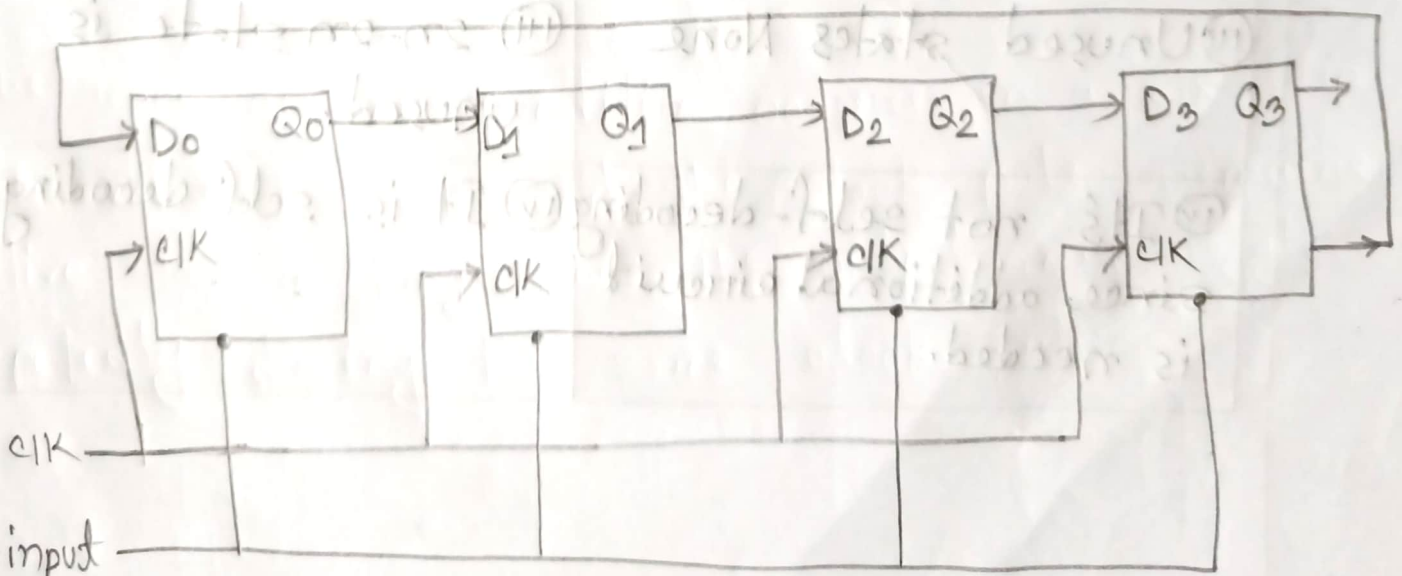


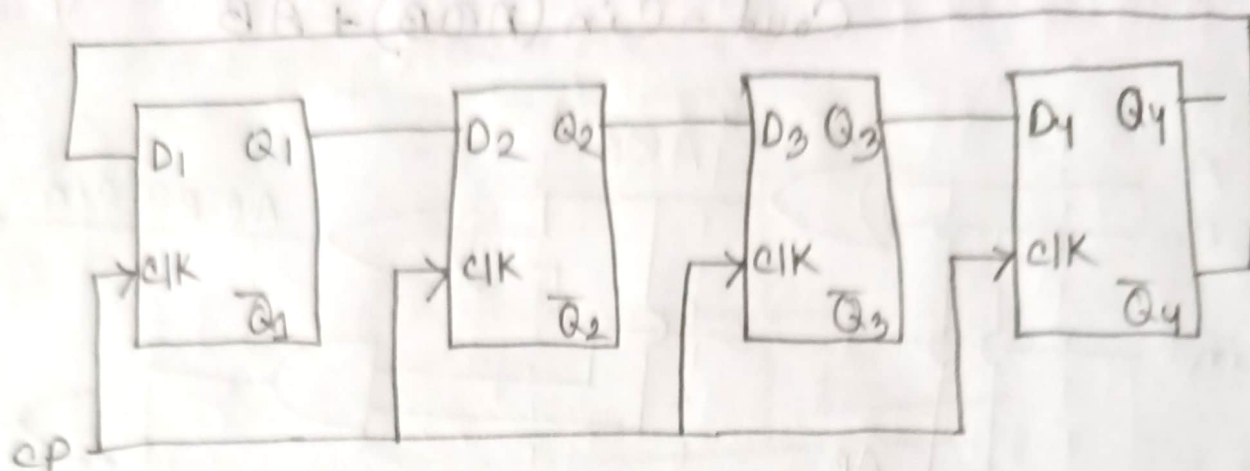
Figure: Two Half Adder on Full Adder.

10. Design Ring Counter / Johnson Counter with state sequences. Compare different Counter.

Ans: Ring Counter:



Johnson Counter :



Ring Counter vs Johnson Counter:

Ring Counter	Johnson Counter
① 'n' flip flops are required to count 'n' state.	① 'n' flip flop are required to count "2n" state.
② It can generate 'n' unique state.	② It can generate '2n' unique state
③ Unused states None	③ 2n-2n state is unused.
④ It's not self-decoding since additional circuit is needed.	④ It is self decoding